CS 3100C: Honors Discrete Math

Lecture 18: Counting Basics

1 The Sum Rule: Choose One of Several Cases

Our first task in counting is to say exactly what one outcome is. We can then split the outcomes into cases, construct them in stages, or match them with objects that are easier to count.

Principle 1 (Sum rule)**.** *If every outcome belongs to exactly one of the disjoint cases $S_1, \ldots, S_m$, then*

$$|S_1 \sqcup \cdots \sqcup S_m| = |S_1| + \cdots + |S_m|.$$

Example 1: a choice from a menu. A cafe offers four kinds of muffins and three kinds of bagels. Choosing one kind gives either one of the four muffin kinds or one of the three bagel kinds. The cases cannot overlap, so there are $\boxed{4+3=7}$ choices.

Example 2: binary strings. How many binary strings of length 4 have *at most one* 1? There is one with no 1s, namely 0000, and four with exactly one 1, determined by its position. Thus there are

$$\boxed{1+4=5}.$$

"At most one" gives two disjoint cases. To add counts, the cases must also include every allowed string.

2 The Product Rule: Make a Sequence of Choices

Principle 2 (Product rule)**.** *Suppose an outcome is built in stages. If stage i always has a_i choices after any valid sequence of earlier choices, and each completed sequence gives a unique outcome, then the number of outcomes is*

$$a_1 a_2 \cdots a_r.$$

Example 1: binary strings. Each of the n positions has two choices, 0 or 1. Therefore

$$\#\{\text{binary strings of length } n\} = \underbrace{2 \cdot 2 \cdots 2}_{n \text{ factors}} = \boxed{2^n}.$$

In particular, length 5 gives $2^5 = 32$ strings.

Example 2: phone numbers. For this lecture, a phone code means a *ten-digit string* using the digits 0 through 9. We allow an initial 0 and ignore real numbering restrictions. Each digit has ten choices, giving

$$\underbrace{10 \cdot 10 \cdots 10}_{10 \text{ factors}} = \boxed{10^{10} = 10{,}000{,}000{,}000}$$

3 The Complement Rule

Sometimes it is easier to count the outcomes that *fail* a condition. If A is a subset of a finite universe U, then

$$\boxed{|A| = |U| - |U \setminus A|}.$$

This follows from the sum rule because A and its complement form a disjoint partition of U.

Example. How many length-5 binary strings have *at least one* 1? There are 2^5 strings in total, and only 00000 has no 1. Hence

$$\boxed{2^5 - 1 = 31}.$$

Adding "strings with a 1 in position i" over all five positions would count a string with several 1s several times.

4 The Bijection Rule

Principle 3 (Bijection rule)**.** *If there is a reversible correspondence between finite sets A and B, then $|A| = |B|$. To show that a correspondence is a bijection, explain how to recover each original object from its image.*

Question. How many length-5 binary strings have an *even* number of 1s?

Let E_5 be the set we want to count. The map appends the parity of the first four bits; the inverse map deletes the fifth bit:

$$\begin{aligned} f : \{0,1\}^4 &\longrightarrow E_5 \qquad\qquad p \equiv b_1 + b_2 + b_3 + b_4 \pmod 2, \\ b_1b_2b_3b_4 &\longmapsto b_1b_2b_3b_4p, \\ g : E_5 &\longrightarrow \{0,1\}^4, \\ b_1b_2b_3b_4b_5 &\longmapsto b_1b_2b_3b_4. \end{aligned}$$

For example, $1011 \overset{f}{\longleftrightarrow_g} 10111$ and $1100 \overset{f}{\longleftrightarrow_g} 11000$. The maps undo each other, so

$$\boxed{|E_5| = |\{0,1\}^4| = 2^4 = 16}.$$

5 Permutations: Order Matters

Phone codes with no repeated digits. How many ten-digit phone codes use no digit twice? The first position has ten choices. Once it is filled, the second has nine choices, then the third has eight, and so on. The tenth position has only one choice. The positions matter: exchanging two digits changes the code. By the product rule, the count is

$$\boxed{10 \cdot 9 \cdots 2 \cdot 1 = 10! = 3{,}628{,}800}.$$

Now consider a four-digit PIN. How many PINs use no digit twice? We make the same choices, but stop after four positions. An initial 0 is allowed, so the count is

$$\boxed{10 \cdot 9 \cdot 8 \cdot 7 = 5{,}040}.$$

If repetition were allowed, every position would again have ten choices and the PIN count would be 10^4.

The general rule. Both examples fill ordered positions from a set of available digits, without reusing a digit. With n distinct objects and r positions, where $0 \le r \le n$, the successive numbers of choices are $n, n-1, \ldots, n-r+1$. Their product is the number of **permutations of r objects chosen from n**:

$$\boxed{P(n,r) = {}_nP_r = \underbrace{n(n-1)\cdots(n-r+1)}_{r \text{ factors}} = \frac{n!}{(n-r)!}}.$$

Thus the phone-code count is $P(10,10)$ and the PIN count is $P(10,4)$. When $r = n$, the formula gives $P(n,n) = n!$; when $r = 0$, it gives $P(n,0) = 1$ for the empty selection.

6 Combinations: Order Does Not Matter

Problem. A class has ten students. How many ways can we appoint a president, a secretary, and a treasurer, with no student holding two roles? How many three-student committees can we form if the members have no special roles?

Officer assignments. The offices are distinct positions, so we use the permutation rule:

$$\boxed{P(10,3) = 10 \cdot 9 \cdot 8 = 720}.$$

Committees. Here we do not care about order. A fixed committee $\{A, B, C\}$ appears once for each assignment of its three members to the three offices. How many repetitions? Exactly $3! = 6$ per committee. Every committee is counted the same number of times, so we divide:

$$\boxed{\frac{P(10,3)}{3!} = \frac{720}{6} = 120}.$$

The general rule. An unordered selection of r distinct objects from n is an r-element subset. Let $C(n,r)$ count these subsets. The key identity is $\boxed{P(n,r) = C(n,r)\, r!}$, read in two directions.

Choose, then order: Choose one subset and permute its r members in $r!$ ways. This produces all ordered selections.

Order, then forget: Start with the $P(n,r)$ ordered selections and forget their order. Each subset appears $r!$ times, so divide by $r!$. Thus

$$\boxed{C(n,r) = {}_nC_r = \binom{n}{r} = \frac{P(n,r)}{r!} = \frac{n!}{r!(n-r)!}}.$$

The notation $\binom{n}{r}$, read "n choose r," is the most common. In particular, the committee count is $\binom{10}{3} = 120$.

Theorem 1 (Symmetry of binomial coefficients)**.** *For* $0 \le r \le n$*,*

$$\boxed{\binom{n}{r} = \binom{n}{n-r}}.$$

Algebraic proof. Since $n - (n - r) = r$,

$$\binom{n}{n-r} = \frac{n!}{(n-r)!\big(n-(n-r)\big)!} = \frac{n!}{(n-r)!r!} = \binom{n}{r}.$$

Intuitive proof. Pair every r-person committee with the set of $n - r$ people *not* on it. Taking the complement twice returns the original committee, so this pairing is a bijection.

7 Problems: Putting the Rules to Work

Problem 1. How many binary strings of length 9 have exactly four 1s?

Solution. Choose the four positions that contain 1; the other five must contain 0. The set of chosen positions determines the string and vice versa, so

$$\boxed{\binom{9}{4} = 126}.$$

Problem 2. How many ten-digit phone codes contain exactly three 5s and exactly two 1s?

Solution. Choose the positions of the 5s, then the positions of the 1s. Each of the five remaining positions may hold any of the eight other digits, independently:

$$\boxed{\binom{10}{3}\binom{7}{2}8^5 = 120 \cdot 21 \cdot 32{,}768 = 82{,}575{,}360}.$$

The remaining digits may match each other, but none can be 1 or 5; otherwise the stated counts would no longer be exact.

Problem 3. A pool has ten seniors and ten juniors. How many four-person teams have *at least one junior*?

Solution. Split by the exact number j of juniors. For each j, choose j juniors and $4 - j$ seniors. The four cases are disjoint and cover every eligible team:

$$\begin{aligned}\#\{\text{eligible teams}\} &= \binom{10}{1}\binom{10}{3} + \binom{10}{2}\binom{10}{2} \\ &\quad + \binom{10}{3}\binom{10}{1} + \binom{10}{4}\binom{10}{0} \\ &= 1{,}200 + 2{,}025 + 1{,}200 + 210 = \boxed{4{,}635}.\end{aligned}$$

8 Permutations with Repeated Symbols

Example: rearrange $AABC$. How many different four-letter strings can we make? Choose the two positions occupied by A; then put B and C in the two remaining positions in either order:

$$\boxed{\binom{4}{2} \cdot 2! = 6 \cdot 2 = 12}.$$

There is another way to see the same count. Temporarily label the two copies A_1 and A_2. All four letters are then distinct, so there are $4! = 24$ permutations. But switching A_1 and A_2 produces the same visible string. Every string is counted exactly $2!$ times, giving $4!/2! = 12$.

The general rule. If a string has n symbols, with multiplicities $m_1, \ldots, m_t$ that sum to n, first label all equal copies so the $n!$ permutations are distinct. Removing the labels counts each visible string $m_1!m_2! \cdots m_t!$ times: each group of equal copies can be permuted without changing the string. Thus the number of distinct arrangements is

$$\boxed{\frac{n!}{m_1!m_2! \cdots m_t!}}.$$

MISSISSIPPI. There are eleven letters: four Is, four Ss, two Ps, and one M. Hence the number of distinct rearrangements is

$$\boxed{\frac{11!}{4!\,4!\,2!\,1!} = 34{,}650}.$$

We can also choose positions in order: four of the eleven for the Is, four of the remaining seven for the Ss, and two of the remaining three for the Ps. The last position holds M, giving $\binom{11}{4}\binom{7}{4}\binom{3}{2} = 34{,}650$.

9 A Final Combinatorial Proof

Theorem 2. *For every integer $k \geq 0$, if $n = 2k$, then $n!/2^k$ is an integer.*

Proof 1: arithmetic. Separate the odd and even factors of $(2k)!$. The even factors are $2, 4, \ldots, 2k$, and each supplies one factor of 2:

$$\begin{aligned}\frac{(2k)!}{2^k} &= \frac{(1 \cdot 3 \cdot 5 \cdots (2k-1))(2 \cdot 4 \cdot 6 \cdots 2k)}{2^k} \\ &= (1 \cdot 3 \cdot 5 \cdots (2k-1))(1 \cdot 2 \cdot 3 \cdots k),\end{aligned}$$

which is a product of integers.

Proof 2: count something. Consider all distinct rearrangements of $a_1a_1a_2a_2 \cdots a_ka_k$, where $a_1, \ldots, a_k$ are different symbols. There are $2k$ positions and each symbol appears twice, so the repeated-symbol formula gives exactly

$$\frac{(2k)!}{\underbrace{2!\,2! \cdots 2!}_{k \text{ factors}}} = \frac{(2k)!}{2^k}$$

distinct strings. A number of strings must be an integer. For $k = 0$, the one empty string gives the same conclusion.